

Virtualisation

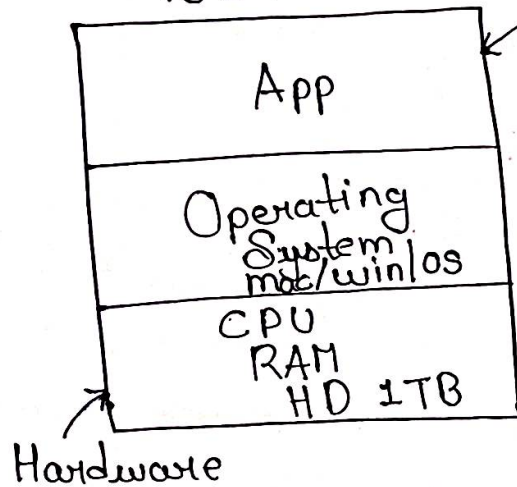
Example: - 1

I got requirement to develop some window applications.

Buying a 1 Server

192.168.1.1

application size
100 mb



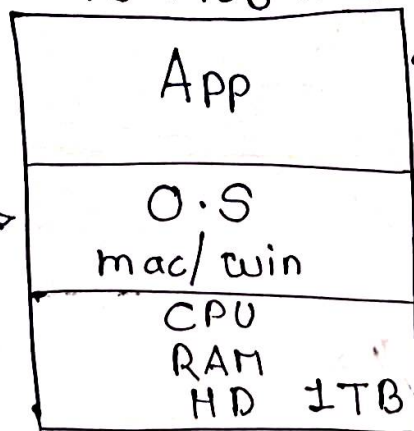
Operating System: - Operating System is interaction Between user and Computer.

Example: 2 one more project

192.168.1.2

application size
300mb

My application
will use 2%
resources only



- In this Scenario we are running 2 Server, for the different-different project But the both Server Configuration, cost are same. if i run in 1st server 100mb application and 2nd server use 300mb, what about my remaining storage. The resources are the wastage. And there are use Concept of Virtualisation.

Virtualisation:- By using virtualization Concept we can create multiple 'Vm'(Server) by sharing the hardware of host machine.

- (On a Customer a Customer demand, we can create a server virtually, any where and any time is call Virtualisation) cloud

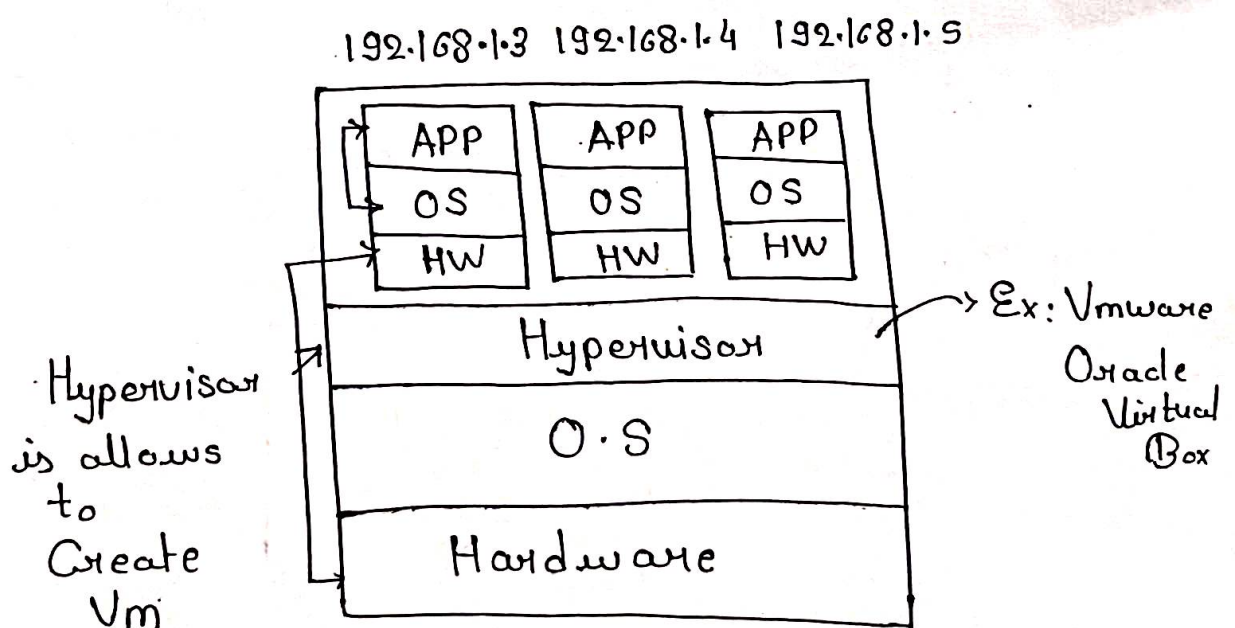


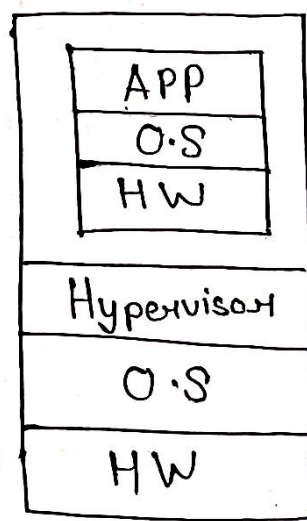
Fig 1.1

Hypervisor is allow to Create a Server Virtuly.

- In Virtually we are creating 3 and more server depending upon a physical hardware configuration. But My base server Harddisk an operating System, Are Corrupted are virtually server are also stoped. There is a problem?

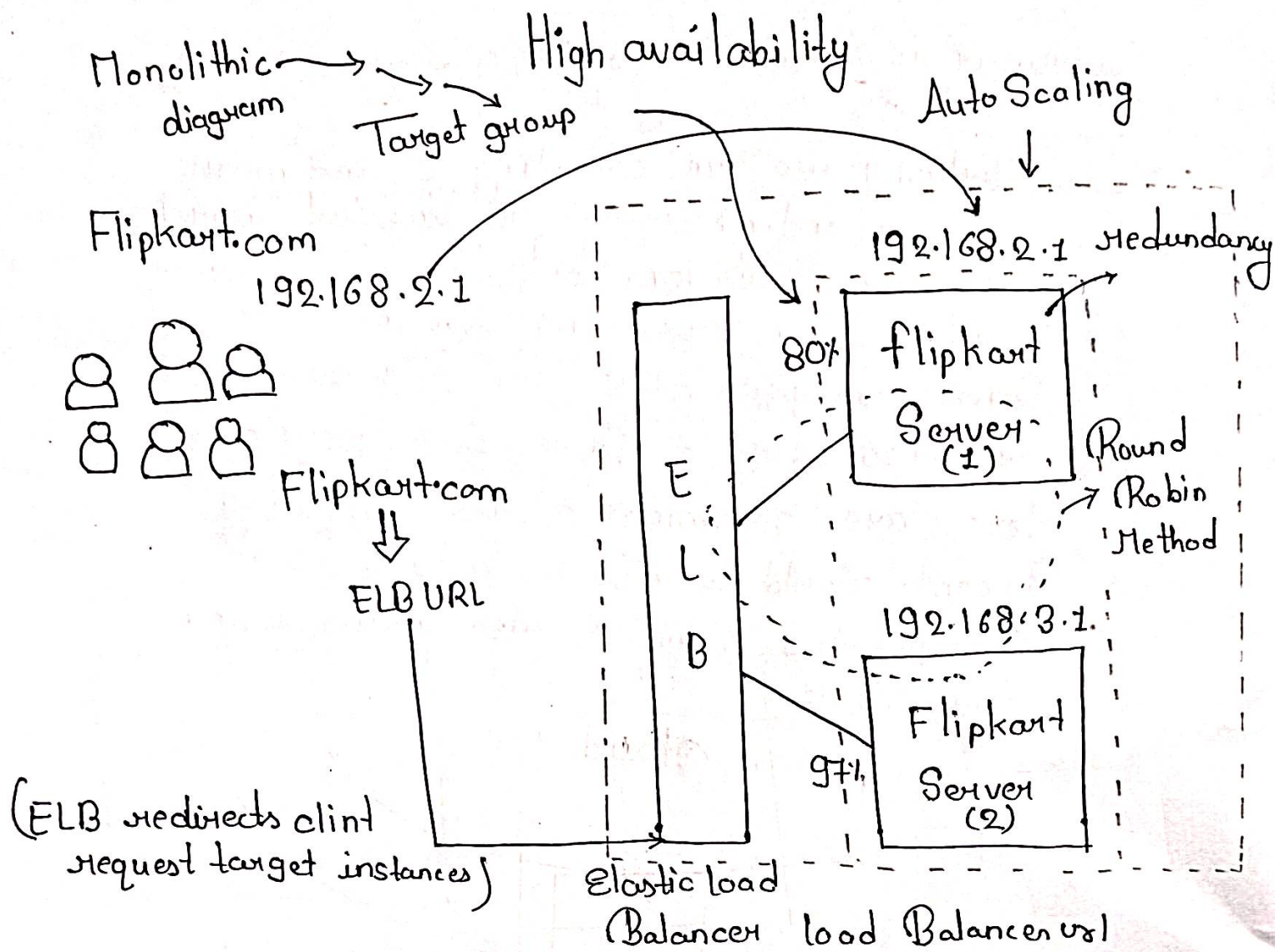
Here are comming cloud Computing Concept. cloud is also use a Hypervisor Concept. And cloud is also using using the Fig 1.1.

Cloud



The server has
to run 24/7

In cloud only our work is to create a Vm and launch that it. In cloud don't care about server down, if one server are down, its automatically running the second server, don't care about billing, rent, server or network Engineer.



In this Scenario, if a client request to a Server(1), then Server is respond to client Ex.

192.168.2.1 and Server ip is 192.168.2.1

Server respond a Same ip, But mo lots of people searching in flipkart Server are down, the we are creating 2nd Server But Second Server ip address are different (192.168.3.1) Server are not interact with Flipkart domain ip.

Because Flipkart domain ip are different and Server2 2 ip different. Question is how to use Same ip with different Server.

Solution is (ELB) Elastic Load Balancer. (Elastic Load Balancer redirects client request to target instances). If Flipkart 1 server is down then load balancer automatically switches to second server.

- ELB always monitors targets (servers = instances)
- ELB supports failover routing (if one server fails it is redirecting to other server)
- if we want consider flipkart application has high availability.

1. Redundancy

2. Monitoring

3. Failover

} Load Balancer distributes the traffic. it will not create any server.

- In technical term we are calling Auto Scaling (Ex. if server CPU load is 80 to 95% it will automatically balance the CPU load.)

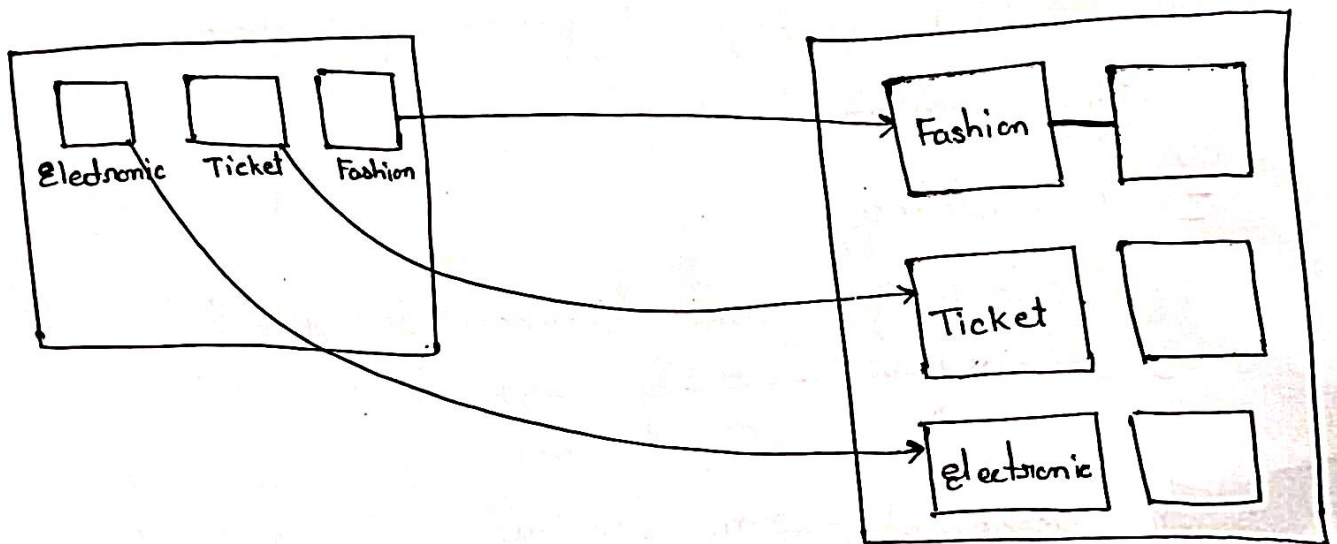
- Auto scaling is creating a server on a same target group. Ex. flipkart.com, one server is fail it will automatically create a second one.

- Question → My auto scaling immediately going to create a server, first and second are already busy with the load

autoscaling immediately create a third server.
then who will tell to auto scaling creating
a third server.

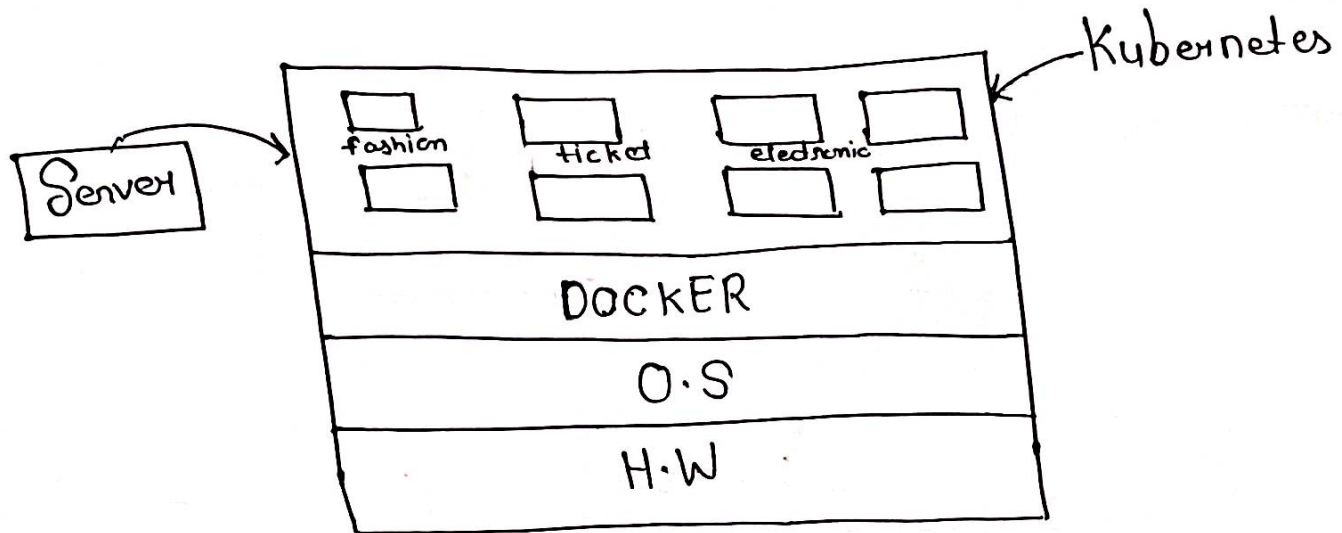
Ans → Auto Scaling dynamically handle
this request, why auto scaling also
monitor your server load.

- This monolithic approach, one server one application individually.

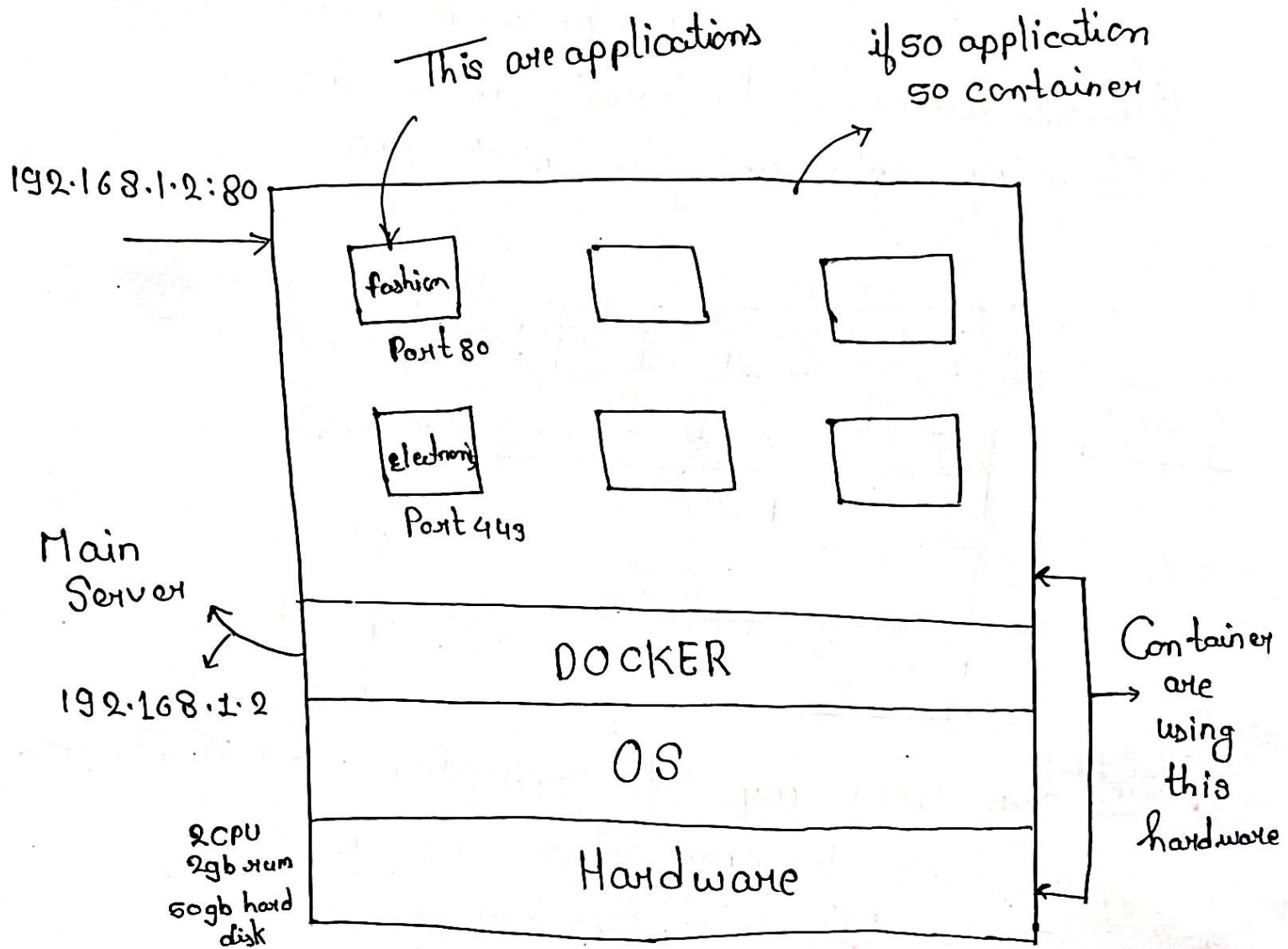


- If I need fashion high availability, then we are creating one more server, then server cost increase, there are used monolithic approach. Ex. using fashion for one server for Ticket one more server if I have 500 application need 500 server means cost increase in one on one server Every server. And it is not easy to monitor 500 server.

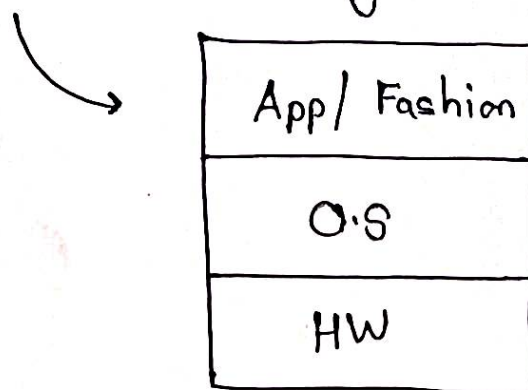
Solution:- That's we are moving with containers Process. No need to create more instances, Creating only one Instance (Server).



- I can host my all application on Container. You don't do any thing that is taking care by Kubernetes. Docker is create a container, Kubernetes also create a Container But he give access Kubernetes Create a load Balancer.
- Docker ~~is~~ will convert your application into image formate first that image is very-very light weight compare to 1cr time less.

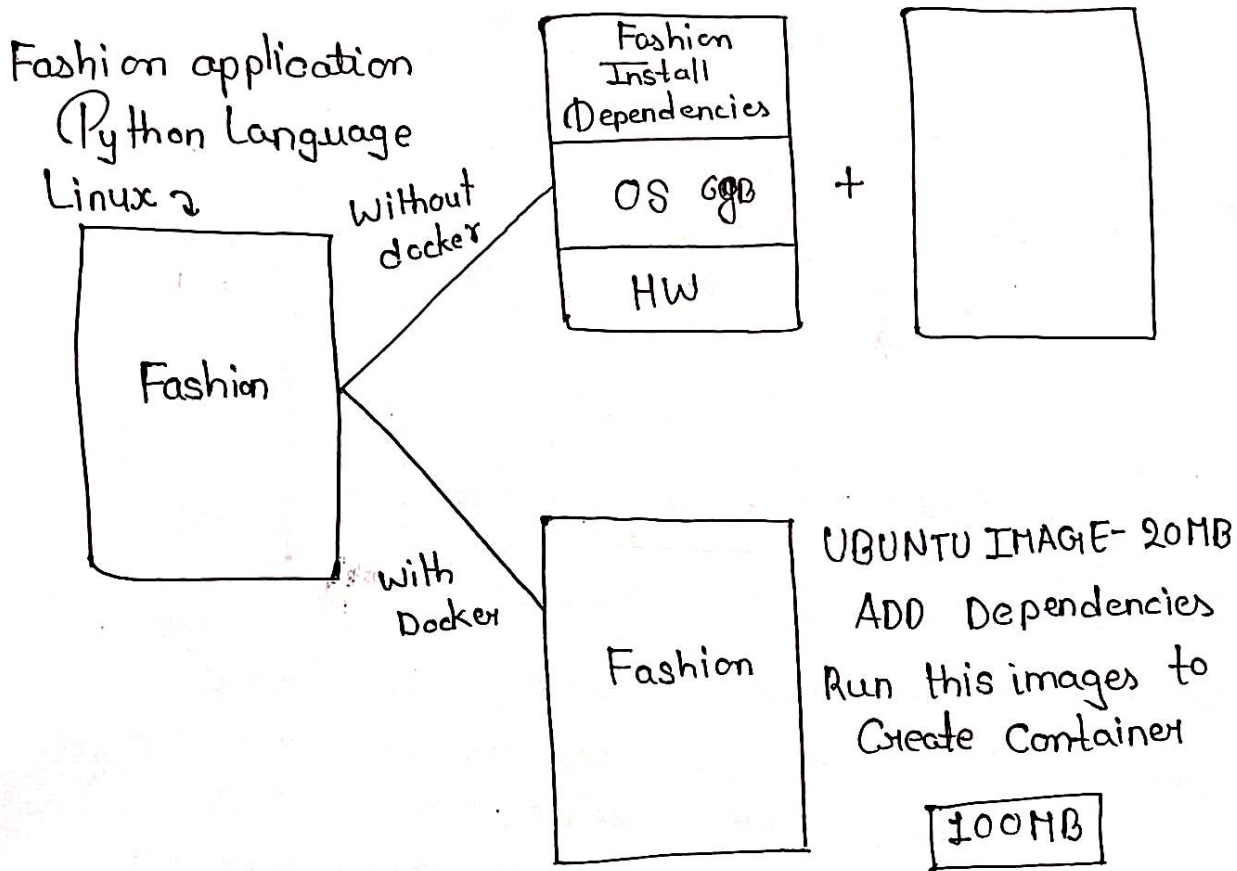


- Docker Container also works like a Server
- Container are works, we only we can give access by enabling the port.
- Container also working on this concept



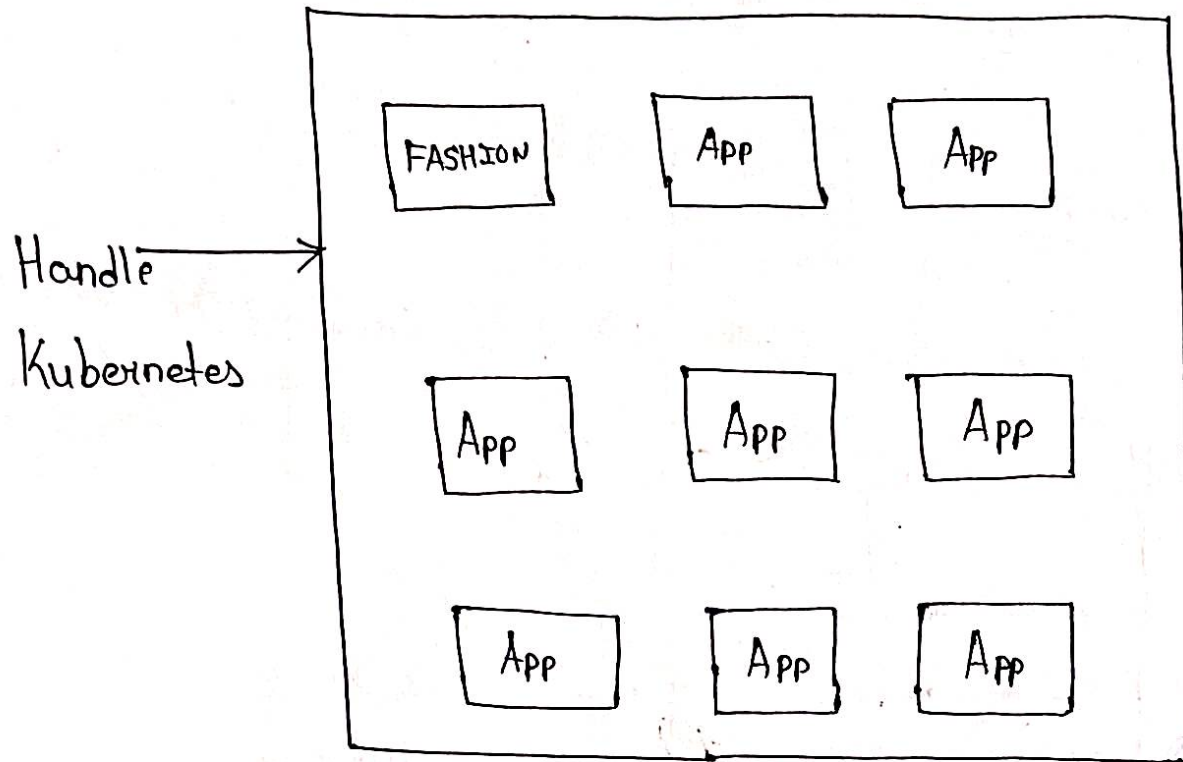
- Docker will give dependencies to run only application

Ex.



In this picture we are understanding the concept of docker, why docker is very useful. I have a Fashion application in python language, without docker what we need to do, Creating a Server, O.S is 6B and install fashion dependencies. If Server load is high we can also create one more Server same Configuration, But there are Server cost is high, more dependencies is waste, more hardware is waste there is the problem without creating docker. with Docker Everything is less 1000 times. Only download the dependencies that are needed. O.S Images is more reduce. There is the are benefit to use docker.

Kubernetes is orchestration of this container



- When docker job is done. then Come Kubernetes, Kubernetes is orchestration tool, How many container are create, how many container are delete, How many container are required Everything is handle Kubernetes:-